

Rational equations



- 1
- | | | | |
|---------------------------|----------------------------|----------------------------|----------------------------|
| a $x = 0.2$ | b $x = 0.45, -0.45$ | c $x = 3.5$ | d $x = 0.32, -0.32$ |
| e $x = 2.62, 9.38$ | f $x = -5$ | g $x = 3.29, -5.17$ | h $x = 2.53$ |

- 2
- a** $x = -3$
- b** No it is not, because it makes two of the denominators in the equation equal to 0.

- 3
- | | |
|---|---------------------------------------|
| a $x = 4$ is not a solution. | b $x = 4$ is a valid solution. |
| c $x = 4$ is not a solution. | d $x = 4$ is not a solution. |
| e $x = 4$ is an extraneous solution. | f $x = 4$ is a valid solution. |

- 4
- | | | | |
|---|-----------------------|---------------------|-----------------------|
| a $x = -\frac{5}{2}$ | b $x = -5$ | c $x = 28$ | d $x = 0$ |
| e $x = 3$ | f $x = 7$ | g $x = 5, 2$ | h $x = -5, -4$ |
| i $x = \frac{3}{2}, \frac{3}{4}$ | j $x = 2, -13$ | | |

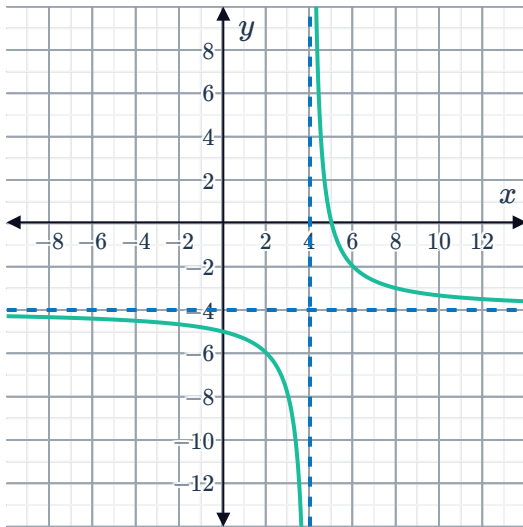
- 5
- | | | | |
|------------------|------------------|----------------------|----------------------|
| a $x = 2$ | b $x = 1$ | c $x = 4, -4$ | d $x = 3, -3$ |
|------------------|------------------|----------------------|----------------------|

- 6 $x = 1, -1, -\frac{i}{3}, \frac{i}{3}$

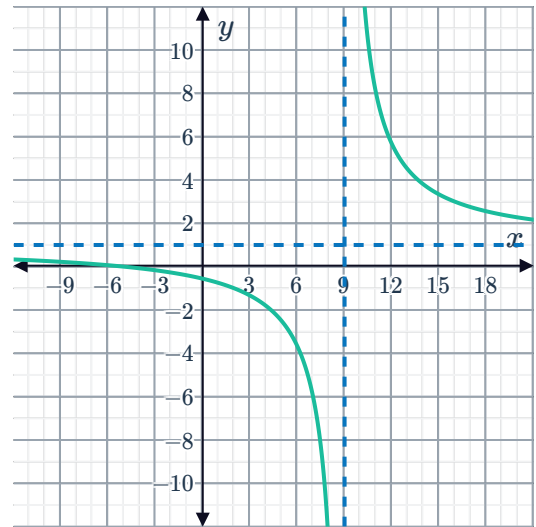
Graphical solutions to rational equations

- 7
- | | | | | | | | | | |
|----------|------------|-------------|--------------|-------------|----------|------------|-------------|--------------|-------------|
| a | i 1 | ii 1 | iii 1 | iv 1 | b | i 2 | ii 1 | iii 0 | iv 1 |
| c | i 1 | ii 1 | iii 1 | iv 2 | d | i 2 | ii 1 | iii 1 | iv 2 |

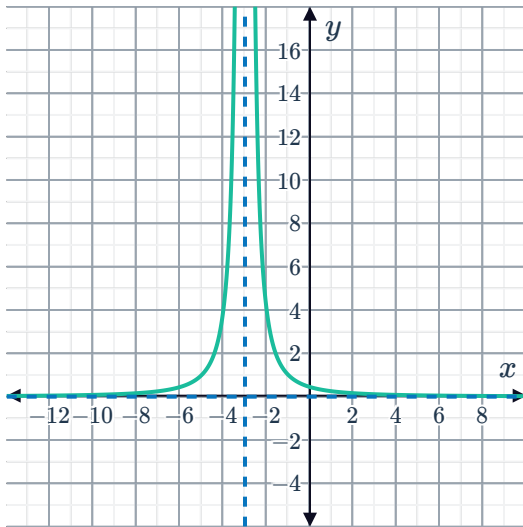
a



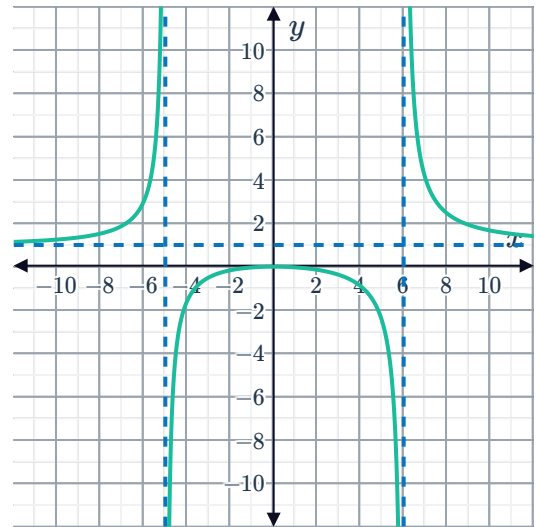
b



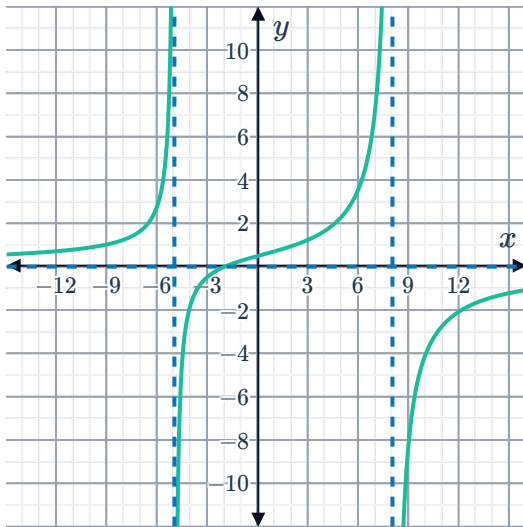
c



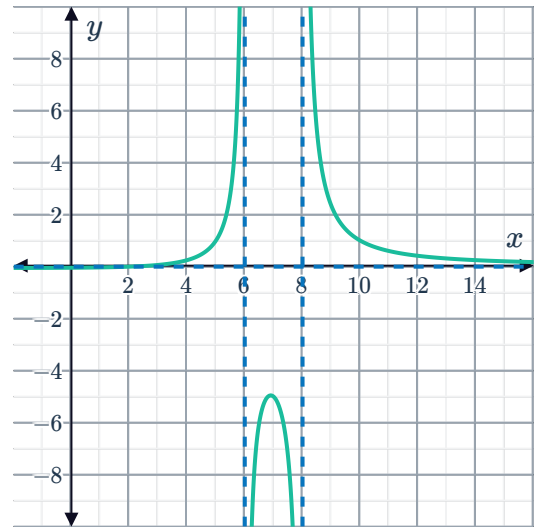
d



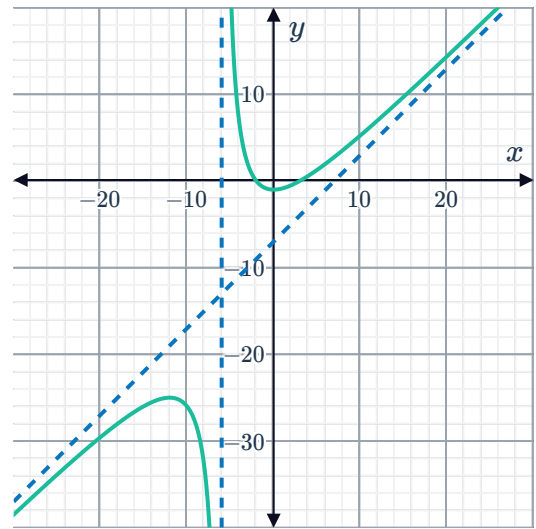
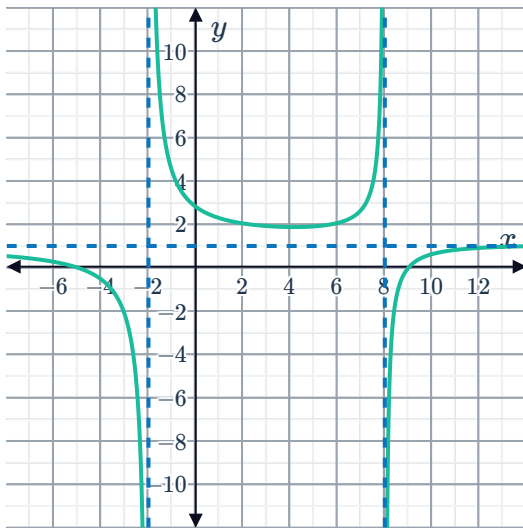
e



f

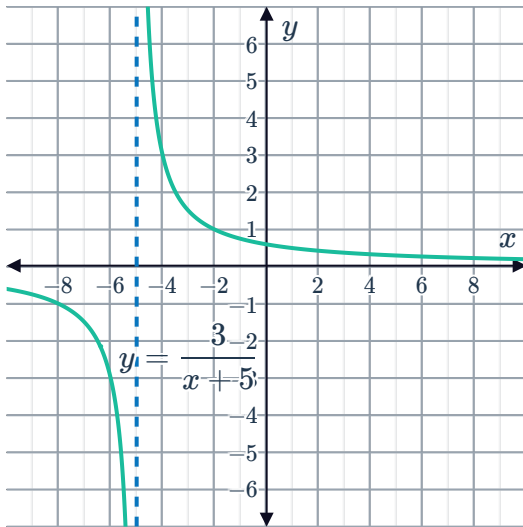


g

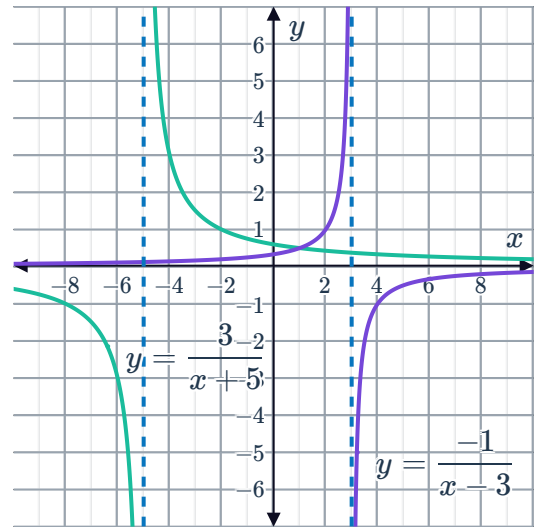


9

a



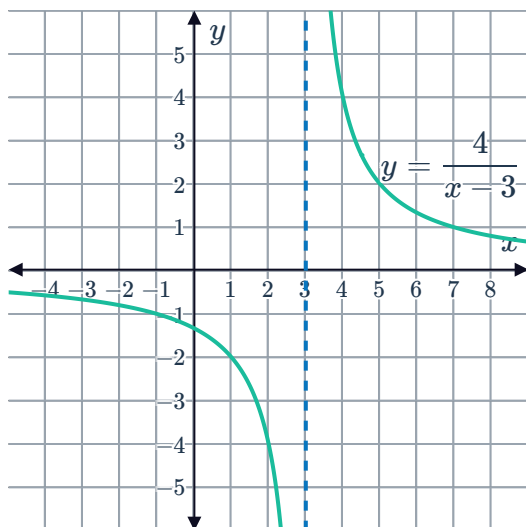
b



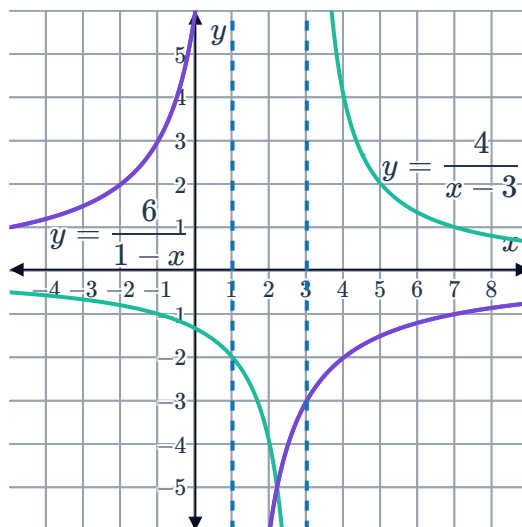
c $x = 1$

10

a



b

c $x = 2.2$